

Melvin Thoompunkal

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PROFESSIONAL SUMMARY

Computer Science sophomore at NJIT (Dean's List) with hands-on experience in software engineering, data engineering, and applied machine learning. Work spans full-stack development, distributed real-time systems, ETL pipeline architecture, and LLM-powered AI systems, with expertise in Python, Java, JavaScript, SQL, TensorFlow, PyTorch, and LangChain. Background includes LLM evaluation at Handshake and customer discovery through NSF I-Corps.

EDUCATION

New Jersey Institute of Technology

Newark, NJ

B.S. Computer Science, Minor in Business, Dean's List

Expected May 2028

Relevant Coursework: Data Structures, C++, Statistics & Probability

EXPERIENCE

IT Intern

June 2026 – August 2026

Sompo Holdings, Inc.

- Architected and deployed an end-to-end agentic AI chatbot for the multinational team, integrating a workflow that scaled to support 500+ multinational users and accelerated policy retrieval time by 75%.
- Engineered rigorous automated evaluation pipelines to continuously benchmark LLM performance, executing 1,000+ test cases that drove a 40% increase in bot accuracy and overall efficiency.
- Led real-time client-side proposal reviews and feedback cycles, rapidly iterating on custom data-formatting scripts to standardize outputs and eliminate 15+ hours of manual workflow bottlenecks per week.

AI Data & Evaluation Fellow

Jan 2026 – Present

Handshake

- Evaluated LLM outputs and conducted structured annotation across text and UI, producing high quality training data.

Entrepreneur Fellow

Feb 2026 – Mar 2026

NSF I-Corps Hub: Northeast Region

- Conducted 20+ structured customer discovery interviews for Team 105: Ocular Insights, applying the Business Model Canvas and lean startup methodology to validate product-market fit.

Marketing Director

Sep 2025 – Present

Business Masterminds, NJIT

- Grew active membership ~35% YoY using SQL-backed audience segmentation models to identify high-conversion outreach targets and align campaigns with executive stakeholders.

PROJECTS

God View

August 2026

Python, FastAPI, React, Kafka, Redis, PostgreSQL/PostGIS, WebSockets

- Architected a scalable real-time global monitoring platform with FastAPI, Kafka, and Redis, ingesting concurrent data streams from GDELT, financial markets, and weather APIs at over 10,000 events per second.
- Engineered asynchronous worker pipelines (aiohttp, asyncpg) and WebSocket broadcasting that reduced external API polling latency by 65% and cut market and weather data retrieval time by 50%.
- Built a spatial PostgreSQL/PostGIS database that accelerated geographic queries by 40%, paired with a React/WebGL frontend rendering satellite imagery and GeoJSON overlays at 60 FPS, plus a Pandas/Scikit-learn pipeline for real-time trend classification.

Graft

Mar 2026 – Apr 2026

Python, JavaScript, Vite, React, Claude API, Sub-Agent Architecture

- Built an AI-powered GitHub repo analysis platform using a multi-model Claude sub-agent architecture with chunked processing, eliminating token overflow while reducing API cost and scan time.
- Engineered a feature extraction pipeline that identifies product features, traces structural implementation paths, and generates portable code bundles as blueprints for reuse.
- Presented a live demo at the Handshake Access 2026 Student Showcase, reaching 4,000+ attendees as a featured student-built AI project.

CERTIFICATIONS

- AWS Fundamentals (AWS)
- Architecting Solutions on AWS (AWS)
- Advanced RAG & Agentic AI (IBM)
- Building with Claude API (Anthropic)

SKILLS

Languages: Python, Java, JavaScript, SQL, C++

ML / AI: TensorFlow, PyTorch, Scikit-learn, LangChain, RAG, ChromaDB, FAISS, Pinecone

Data Engineering: ETL Pipelines, Apache Spark, Databricks, Azure, PostgreSQL/PostGIS, Kafka, Redis, Power BI, Pandas

Tools & Frameworks: FastAPI, Flask, Streamlit, Git, WebSockets, Matplotlib, Plotly, Google Sheets API